Abdelrahman Khalifa

Embedded Software Engineer | C/C++, RTOS, Safety-Critical Systems

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SUMMARY

Embedded software engineer writing C firmware on an RTOS for real machines: owns the Zephyr RTOS firmware of EKSON's pneumatic motion simulators, with safety stops and a watchdog. Automotive background from an AUTOSAR-based V2X graduation project (bootloader with image verification and rollback), AUTOSAR drivers and a FreeRTOS controller with fault diagnostics. Merged contributions to Zephyr RTOS and The Linux Foundation's OpenPrinting. Works in C, C++ and Python; based in Cairo.

EXPERIENCE

Embedded Software Engineer | **EKSON** | **Aug 2025 – Present**

Software for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Wrote the ESP32-S3 firmware in C and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS for more deterministic timing and devicetree-based drivers.
- Implemented closed-loop PID control of the pneumatic valves with feedback from an MPU6050 IMU over I2C, plus safety functions: emergency stop, motion limits and a watchdog.
- Developed the desktop control app (UART protocol to the device) and cut its CPU use and latency by about 40% (measured) with asyncio; built firmware-update delivery and GitHub Actions CI.

Google Summer of Code Contributor | **The Linux Foundation (OpenPrinting)** | **May 2026 – Oct 2026**

Open-source project: Printers page for COSMIC Settings, in Rust | github.com/Abd002/GSOC-2026

- Investigated and fixed crashes and a leaked client in libcups' C code, and isolated the native backend in a separate service so a crash cannot take down the app; **11 PRs merged upstream** (list).
- Built CI that integration-tests the code against real CUPS and Avahi services on x86_64 and aarch64.

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | **Dec 2024 – Jan 2025**

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OPEN SOURCE PROJECTS

- **Zephyr RTOS** (Mar 2025): implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules, reviewed and merged (issue #86977).

PROJECTS

- **AUTOSAR-Based V2X Vehicle** (graduation project, team; STM32, ESP32): owned the firmware update path. The image is sent over UART with UDS services (request download, transfer data, transfer exit); the STM32 bootloader verifies it with the hardware CRC unit before jumping to it and keeps two banks with rollback. GitHub
- **Seat Heater Control System** (Tiva C, FreeRTOS): two-seat real-time controller; event groups and periodic tasks, a mutex-protected shared UART display, sensor-failure detection logged to memory with a fault LED, and run-time measurement of CPU load and task execution times. GitHub
- **AUTOSAR PORT and GPIO drivers** (C, TM4C): MCAL drivers developed and tested against the AUTOSAR specification.
- **Door Locker Security System** (C, two ATmega32): layered HMI and control ECUs communicating over UART; password check, EEPROM, buzzer alarm and PWM-driven motor. GitHub
- **Traffic Light Controller** (Verilog, FPGA): finite state machine with debounced inputs and timing modules. GitHub

TECHNICAL SKILLS

Languages: C, Modern C++, Python, Rust, Bash, Verilog

Embedded & RTOS: Zephyr, FreeRTOS, OSEK/VDX (Trampoline), AUTOSAR (MCAL), bootloaders, UDS; ARM Cortex-M (STM32, TM4C), ESP32-S3, AVR

Interfaces: UART, I2C, SPI, CAN; TCP/IP sockets

Testing & tools: unit and integration testing, CI (GitHub Actions), fault diagnostics, Git, CMake, Make, debugging, Linux

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | **2020 – 2025**

- Grade: Very Good with Honors (84.6%). Codeforces Expert; 4-time ECPC finalist (2021–2024).